Group Project Checkpoint

Predicting 30-Day Customer Repeat Purchases Using Behavioural and Semantic Product Features

ELEC 3544 | Group 1 | Clara Natalie Surjadjajadi | Student ID 3036029686 | 7 November 2026

1. Group progress

Project question. Can a customer's purchases in the previous 90 days predict whether they buy again in the next 30 days? We compare basic RFM features, extra behaviour features and MiniLM product-text features, using logistic regression and gradient boosting. The main metric is PR-AUC.

Dataset. UCI Online Retail (ID 352): 541,909 invoice lines, Dec 2010 to Dec 2011. The audit found 5,268 duplicate rows and 135,080 rows with no CustomerID. After cleaning, 391,153 merchandise lines are used, and postage, fee and voucher lines are excluded.

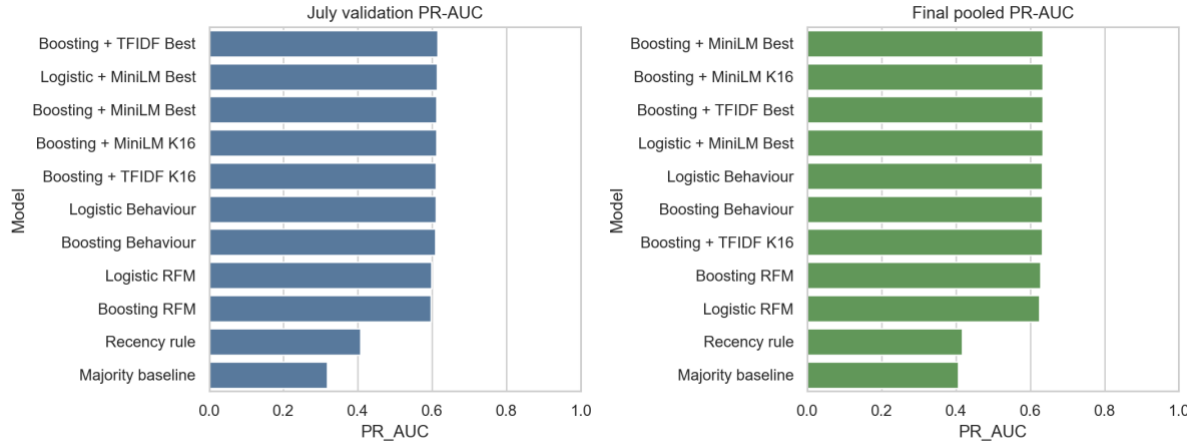
Work completed. An executed notebook (27 code cells) that cleans the data, builds 16,123 customer-month rows (Mar to Oct 2011), creates the features and TF-IDF / MiniLM product clusters, and tests nine models plus two baselines. Training uses Mar to Jun, July selects the model, and Aug to Oct is the final test, so no test month influences any choice. The pipeline is complete and executed. The remaining work is verification, final figures, the report and the slides.

Preliminary findings. Every model beats the baselines (Table 1). Extra behaviour features help logistic regression (+0.008 PR-AUC, interval above zero) but not clearly boosting. The July-selected model, boosting + LSA-TF-IDF, scores 0.634 on the final test with lift 1.77 at the top 20%. Purchase rhythm (ActivePurchaseDays) is by far the most important feature.

Unsuccessful results. Text features gave no clear gain: MiniLM adds +0.001 and TF-IDF +0.001 PR-AUC over behaviour-only boosting, with intervals that include zero. We report this as a negative result. Our first TF-IDF clustering was too weak to compare fairly (cosine silhouette 0.015, one cluster holding about 68% of products), so we replaced it with LSA-TF-IDF (0.145, largest cluster about 25%). MiniLM clusters score lower (0.075 at K=20, and 0.067 against 0.123 at the matched K=16).

Table 1. PR-AUC by model (July = validation month, Aug-Oct = pooled final test)

Model or baseline	July PR-AUC	Aug-Oct PR-AUC
Majority baseline	0.317	0.406
Recency rule	0.407	0.418
Logistic RFM	0.598	0.625
Boosting, behaviour	0.608	0.633
Boosting + LSA-TFIDF	0.615	0.634
Boosting + MiniLM	0.612	0.634



Evidence: *Table 1 and the figure (notebook_outputs/model_comparison.png, left two panels) come from Online_Retail_Repeat_Purchase_Project.ipynb (validation_results.csv, final_pooled_results.csv). The top learned models are within about 0.002 PR-AUC of each other.*

2. Individual contributions

Name	Specific work completed	Evidence / output	Approximate hours spent	Next task
Clara Natalie Surjadjajadi (me)	Stages 10-13 and integration. Extended the notebook with uncertainty, ablation and sensitivity analyses (PR #1). Reran the notebook myself, reviewed the feature stages, July model selection and the Aug-Oct test, and checked the numbers in this checkpoint against the notebook outputs. Report integration remains.	Executed notebook (Stages 10-13), customer_snapshot_features.csv, validation_results.csv, rolling_monthly_results.csv, paired_model_differences.csv	19	Verify the pipeline section against my rerun, merge and review the final report
Stacia Benita	Stages 1-3: data audit, cleaning and invoice table. Reviewed the EDA, merchandise/charge split, revenue audit and duplicate check (removing duplicates changes 0.27% of valid revenue and no monthly label). Submitted her own checkpoint.	eda_data_quality.csv, eda_monthly_activity.csv, invoice_revenue_audit.csv, cleaning_audit.csv	18	Independently rerun Stages 1-3, verify duplicate and charge-line decisions
Devano Jonathan	Stages 8-9 and October error analysis. Replaced raw sparse TF-IDF with LSA, added the matched K=16 comparison, checked clusters and errors, and directed the notebook repair.	cluster_quality.csv, cluster_matched_k_quality.csv, product_cluster_examples.csv, error_analysis_summary.csv	20	Accept the cluster descriptions, review final wording

AI assistance and how I checked it. I used AI to assist me and to write code: the pipeline notebook was generated with Codex from our written specification, and I used Claude Code to help read the project files and draft this checkpoint. I did not simply trust the code or its output. I verified the results myself in four ways: (1) I compared each PR-AUC value in Table 1 with validation_results.csv and final_pooled_results.csv, (2) I compared the data counts (541,909 lines, 5,268 duplicates, 135,080 missing IDs, 391,153 merchandise lines) with the notebook's printed audit and cleaning tables, (3) I read the split code to confirm that training uses Mar to Jun, July is only used for selection and Aug to Oct is only used for testing, so no future data leaks into the features, and (4) I checked that the claims match the evidence, for example that the text-feature gains are reported as not distinguishable from zero because their intervals include zero. I also reran the notebook myself. The interpretation and wording are my responsibility.

3. Next steps and blockers

Remaining major task	Owner	Target date
Submit individual two-page checkpoint with confirmed hours	Each member	7 Nov 2026
Reconcile sensitivity and cluster-count results, finalise figures, keep the iteration log	Clara, Devano, Stacia	9-14 Nov 2026
Verify the pipeline section against my notebook rerun	Clara	9-14 Nov 2026
Independently rerun Stages 1-3, verify duplicate and charge-line decisions	Stacia	9-14 Nov 2026
Accept cluster descriptions, review final wording	Devano	9-14 Nov 2026
Merge report sections with one consistent argument and caveat set	All	15-20 Nov 2026
Build slides, rehearse, final numerical consistency check	All	21-25 Nov 2026

Blockers and support needed. Each member must still confirm their own hours and verify their assigned evidence. The text-feature gains are tiny (under 0.002 PR-AUC), so the report must not claim a winner. Cluster labels still need human acceptance.